

Problem: For what real values of x is

$$\sqrt{x + \sqrt{2x - 1}} + \sqrt{x - \sqrt{2x - 1}} = A$$

given

1. $A = \sqrt{2}$
2. $A = 1$
3. $A = 2$

where only non-negative real numbers are admitted for square roots? (Source: 1959 IMO)

Solution: We can simplify this equation by squaring both sides.

$$\begin{aligned} & \sqrt{x + \sqrt{2x - 1}} + \sqrt{x - \sqrt{2x - 1}} = A \\ \left(\sqrt{x + \sqrt{2x - 1}} + \sqrt{x - \sqrt{2x - 1}} \right) \left(\sqrt{x + \sqrt{2x - 1}} + \sqrt{x - \sqrt{2x - 1}} \right) &= A^2 \\ x + \sqrt{2x - 1} + \sqrt{x^2 - (2x - 1)} + \sqrt{x^2 - (2x - 1)} + x - \sqrt{2x - 1} &= A^2 \\ 2x + 2\sqrt{x^2 - (2x - 1)} &= A^2 \\ 2x + 2\sqrt{x^2 - 2x + 1} &= A^2 \\ 2x + 2\sqrt{(x - 1)^2} &= A^2 \\ 2x + 2(x - 1) &= A^2 \\ 4x - 2 &= A^2 \\ 4x &= A^2 + 2 \\ x &= \frac{A^2 + 2}{4} \end{aligned}$$

When $A = \sqrt{2}$, we have $x = 1$. When $A = 2$, we have $x = \frac{3}{2}$.

It looks like $A = 1 \implies x = \frac{3}{4}$. But plugging $x = \frac{3}{4}$ into the original equation does not give us $A = 1$. So I think there is no solution for $A = 1$.

Thus the solutions are $x = \left\{ 1, \frac{3}{2} \right\}$.